

Lorenzo SALA, PhD

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Career

- **Junior Research Scientist (CRCN)** at MaIAGE, INRAE Jouy-en-Josas, Université Paris-Saclay and MUSCA, INRIA Saclay Ile-de-France. 2023 - now.
- **Post-doc** under the supervision of I. Vignon-Clementel at SIMBIOTX, INRIA Saclay, Ile-de-France. 2020 - 2022.
- **Research Associate** under the supervision of P. Degond at Imperial College London. 2019 - 2020.
- **PhD in Applied Mathematics** under the supervision of C. Prud'Homme, G. Guidoboni, M. Szopos at the Université de Strasbourg. 2016 - 2019.
- **Research Engineer** at the Université de Strasbourg. May - September 2016.
- **MSc in Computational Science and Engineering** at Politecnico di Milano. 2013 - 2016.
Master internship at Micron Electronics (8 months) under the supervision of R. Sacco, A.G. Mauri, G. Guidoboni.

Awards

- **VPHi-InSilicoTrials PhD Thesis Award in In Silico Medicine for potential application in industrial R&D.** 2020.
- **Best PhD Thesis Award - Prize of the Research Commission of the University of Strasbourg.** 2020.
- **Best Poster Award** - University of Strasbourg Doctorate School in Mathematics, Engineering and Computer Science. 2017.

Organization scientific events

- Scientific committee and organizer of a special session at *18th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering (CMBBE 2023)*.
- CEMRACS 2023¹ project organizer
- Organizer of minisymposia at *SIAM Conference on Computational Science and Engineering (CSE23)*, *SMAI 2025*, and World Congress of Computational Mechanics (WCCM) 2026.
- Organizer of workshop at the *American Institute of Mathematics*, 2025.

Supervision

- X. Amorós Gabarrón. PhD. *Modelling the dynamics of plant immune responses with physics-assisted artificial intelligence*. Co-supervision with S. Bottini (Institut Sophia Agrobiotech, Sophia-Antipolis), R. Duvigneau (Inria Sophia-Antipolis), A. Garcia-Molina (Centre for Research in Agricultural Genomics, Barcelona). 2026 - now.
- E. Pastremoli. PhD. *Towards a digital twin of the gut microbiota: a multidisciplinary approach for an in-depth understanding of composition, function and interaction with the host*. Co-supervision with B. Laroche (INRAE Jouy-en-Josas). 2023-2026.
- J. Kowalski. PhD. *Whole body vascular transport and pharmacokinetics models: application to imaging, in particular of liver*. Co-supervision with I. Vignon-Clementel (INRIA Saclay Ile-de-France). 2022 - 2026.
- M. Haghebaert. PhD. *Tools and methods for dynamics modeling of complex microbial ecosystems from experimental temporal observations: application to the dynamics of intestinal microbiota*. 3rd year co-supervision with B. Laroche (INRAE Jouy-en-Josas). 2023.
- G. Lacour. Post-doc. *Physics-informed modelling and data integration for microbial community dynamics (CULTISSIMO project)*. Co-supervision with B. Laroche (INRAE Jouy-en-Josas). 2025-2026.
- L. Perrin, T. Saigre, P.J. Hossie. CEMRACS project (5 weeks hackathon). *Estimation of interactions in microbial communities via a neural network-based generalized smoothing algorithm*. CIRM, Marseille. 2023.
- *Students internships*: A. Pelella, M. Baggi, B. Allieux (2026, INRAE Jouy-en-Josas), V. Cagnazzi (2026, Université Paris Cité), V. Schmitt, M. Denoual, O. Gawas (2025, INRAE Jouy-en-Josas), F. Hughes (2024, University of Maine), A. Dechamps, Y. Jouanaud (2024, INRAE Jouy-en-Josas), K. Hezez (2024, Sorbonne Université), K. Lyons (2023, Université Paris Cité), J. Kowalski (2022, INRIA), L. Thiebaud (2021, INRIA), A. Walczak, G. Kim-J. Liu-Z. Sun-B. Xiao. (2020, Imperial College), N. Marazzi (2019, U. Missouri).

¹<http://smai.emath.fr/cemracs/cemracs23/index.html>

Significant publications

- Hossie, P.J., Laroche, B., Malou, T., Perrin, L., Saigre, T., **Sala, L.** (2025). *Surrogate modeling of interactions in microbial communities through Physics-Informed Neural Networks*. ESAIM: Proceedings and Survey.
- Hanna, J. M., Varsos, P., Kowalski, J., **Sala, L.**, et al. (2025). *A comparative analysis of metamodels for 0D cardiovascular models, and pipeline for sensitivity analysis, parameter estimation, and uncertainty quantification*. Computers in Biology and Medicine, 193, 110381.
- **Sala, L.**, Zaid, M., Hughes, F., Szopos, M., Huxley, V. H. , et al.. (2026) *Eye2Heart: A reduced mathematical model bridging cardiovascular and ocular hemodynamics*. Mathematical Biosciences and Engineering, 2026, 23 (2), pp.421 - 448.
- Haghebaert, M., Laroche, B., **Sala, L.**, Mondot, S., Doré, J. (2024). *A mechanistic modelling approach of the host-microbiota interactions to investigate beneficial symbiotic resilience in the human gut*. Journal of the Royal Society Interface, 21(215), 20230756.
- **Sala, L.**, Prud’Homme, C., Guidoboni, G., Szopos, M., & Harris, A. (2024). *The ocular mathematical virtual simulator: A validated multiscale model for hemodynamics and biomechanics in the human eye*. International journal for numerical methods in biomedical engineering, 40(2), e3791.